

CLASS -10 (2025-26)

INTRODUCTION TO OBJECT ORIENTED PROGRAMMING CONCEPT

CHAPTER 1

Assignments:-

A. Tick (✓) the correct answer.

Which of the following is not a principle of OOP?

Answer: d. Class

Which of the following are the advantages of polymorphism?

Answer: c. Both a and b

(a. Codes can be reused, b. It makes the program run faster)

Which of the following principles of OOP allows the concept of reusability?

Answer: a. Polymorphism

Which of the following is the main element of object-oriented programming?

Answer: b. Objects

Procedural programming splits the programming code into small parts called

Answer: a. Procedures

B. Fill in the blanks.

A **paradigm** is a way of programming.

Procedure-Oriented Programming has global data sharing of functions.

A **low-level** language is a programming language that is machine-dependent.

The concept of **inheritance** is a good feature for avoiding data redundancy.

Java is an example of **object-oriented** programming language.

C. Short Answer Type Questions

What is the use of inheritance?

Answer: Inheritance allows a class to acquire the properties and behaviors (methods) of another class, promoting code reusability and reducing redundancy.

What does POP stand for?

Answer: POP stands for **Procedure-Oriented Programming**.

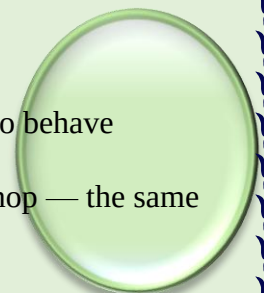
Define polymorphism with a real-life example.

Answer: Polymorphism means "many forms." It allows the same function or method to behave differently based on the object calling it.

Example: A person can be a teacher in school, a parent at home, and a customer in a shop — the same individual exhibiting different behaviors in different situations.

What are the disadvantages of Procedure-Oriented Programming?

Answer:



Difficult to manage large codebases.
 Poor data security due to global data access.
 Limited code reusability.
 Lack of scalability and maintainability.

What are the differences between POP and OOP?

Answer:

POP (Procedure-Oriented Programming)	OOP (Object-Oriented Programming)
Follows a top-down approach	Follows a bottom-up approach
Focuses on procedures or functions	Focuses on objects and classes
Data is global and shared among all functions	Data is encapsulated inside objects
Does not support inheritance or polymorphism	Supports inheritance and polymorphism
Low data security due to global access	High data security through encapsulation and access control
Limited code reusability	High code reusability through inheritance and modular design
Harder to maintain and modify as the program grows	Easier to maintain and extend due to modular structure
Examples: C, Pascal	Examples: Java, C++, Python (OOP features)

- **POP** follows a top-down approach; **OOP** follows a bottom-up approach.
- In **POP**, data is global; in **OOP**, data is encapsulated.
- **POP** focuses on procedures/functions; **OOP** focuses on objects.
- **OOP** supports inheritance and polymorphism; **POP** does not.
- **OOP** provides better data security and reusability than **POP**.

1. Difference between Abstraction and Encapsulation

Abstraction	Encapsulation
Hides implementation details and shows only the essential features to the user.	Binds data and functions into a single unit (class) and hides internal data.
Focuses on what an object does.	Focuses on how data is protected and maintained.
Achieved using abstract classes or interfaces .	Achieved using classes and access specifiers (private, public, protected).
Example: Driving a car without knowing how the engine works.	Example: Data members are private, and access is provided via public methods.
Promotes simplicity .	Promotes security .

2. Difference between Encapsulation and Inheritance

Encapsulation	Inheritance
Encapsulation is the process of binding data and methods that operate on the data into a single unit.	Inheritance is the mechanism by which one class acquires properties and behaviors of another class.
It helps in protecting data from unauthorized access.	It helps in code reusability and creating a hierarchical relationship.
Achieved using access modifiers and classes .	Achieved using extends keyword in Java.
Example: Private data members with public getter and setter methods.	Example: A <code>Car</code> class inherits from a <code>Vehicle</code> class.
Promotes data hiding .	Promotes reusability and extensibility .

3. Difference between Inheritance and Polymorphism

Inheritance	Polymorphism
Enables a new class to inherit properties and behaviors from an existing class.	Allows methods to perform differently based on the object calling them.
Promotes code reusability .	Promotes flexibility and dynamic behavior in code.
Achieved using the extends keyword in Java.	Achieved using method overloading or overriding .
Example: <code>Dog</code> class inherits from <code>Animal</code> class.	Example: <code>draw()</code> method behaves differently for <code>Circle</code> and <code>Rectangle</code> .
Relationship is "is-a" (e.g., Dog is an Animal).	Relationship is "behaves differently" for same interface.

4. Difference between Abstraction and Inheritance

Abstraction	Inheritance
Hides implementation details and shows only essential features.	Allows one class to reuse code from another class.
Focuses on what to do, not how to do it.	Focuses on building a hierarchical relationship .
Achieved using abstract classes and interfaces .	Achieved using the extends keyword in Java.
Example: Interface <code>Shape</code> has <code>draw()</code> method with no body.	<code>Circle</code> class inherits <code>draw()</code> from <code>Shape</code> .
Promotes simplicity and clarity .	Promotes code reusability and organization .

5. Difference between Encapsulation and Polymorphism

Encapsulation	Polymorphism
Binds data and methods into a single unit and restricts access.	Allows one interface to be used for different implementations .
Achieved using classes and access specifiers .	Achieved through method overloading and overriding .
Focuses on data hiding and security .	Focuses on dynamic behavior and flexibility .
Example: Private variables with getter/setter methods.	Example: <code>print()</code> method works for integers, strings, etc.
Promotes security and control .	Promotes extensibility and readability .

Multiple Choice Questions (MCQs) with Answers

1. Which feature of OOP binds data and functions that operate on the data into a single unit?

- a) Inheritance
- b) Polymorphism
- c) Abstraction
- d) Encapsulation

✓ **Answer:** d) Encapsulation

2. Which OOP principle allows a function or method to behave differently based on the object?

- a) Inheritance
- b) Polymorphism
- c) Encapsulation
- d) Abstraction

✓ **Answer:** b) Polymorphism

3 Which of the following languages is primarily based on OOP?

- a) C
- b) Assembly
- c) Java
- d) Pascal

✓ **Answer:** c) Java

4 In Procedure-Oriented Programming, data is mainly:

- a) Hidden inside classes
- b) Shared globally among functions
- c) Accessed only by objects
- d) Managed by constructors

✓ **Answer:** b) Shared globally among functions

5 Which of the following is not a benefit of Object-Oriented Programming?

- a) Code Reusability
- b) Better Data Security
- c) Procedural Flow Control
- d) Easier Maintenance

✓ **Answer:** c) Procedural Flow Control



6. Which of the following is an example of a low-level language?

- a) Java
- b) C++
- c) Assembly
- d) Python

✓ **Answer:** c) Assembly

7. The feature of OOP that hides unnecessary details from the user is called:

- a) Polymorphism
- b) Encapsulation
- c) Inheritance
- d) Abstraction

✓ **Answer:** d) Abstraction

8. Which of the following is not a valid concept in OOP?

- a) Modularity
- b) Global Variables
- c) Polymorphism
- d) Inheritance

✓ **Answer:** b) Global Variables

9. The class in OOP serves as a:

- a) Blueprint for objects
- b) Function library
- c) Database
- d) Data entry form

✓ **Answer:** a) Blueprint for objects

10. Which programming approach is best suited for large and complex applications?

- a) Procedural Programming
- b) Structured Programming
- c) Object-Oriented Programming
- d) Linear Programming

✓ **Answer:** c) Object-Oriented Programming

Assertion and Reason Questions with Options

1.

Assertion (A): Object-Oriented Programming provides better data security than Procedure-Oriented Programming.

Reason (R): OOP uses encapsulation to restrict direct access to data.

Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true, but R is false
- d) A is false, but R is true

✓ **Answer:** a) Both A and R are true, and R is the correct explanation of A



2.

Assertion (A): In POP, code reusability is high due to the use of global variables.

Reason (R): Global variables can be accessed by any function in POP.

Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is false, but R is true
- d) A is true, but R is false

✓ **Answer:** c) A is false, but R is true

3.

Assertion (A): Inheritance helps reduce code redundancy.

Reason (R): Inheritance allows a class to reuse the properties of another class.

Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true, but R is false
- d) A is false, but R is true

✓ **Answer:** a) Both A and R are true, and R is the correct explanation of A

4.

Assertion (A): Polymorphism allows the same method to behave differently based on context.

Reason (R): It helps in defining multiple methods with the same name but different parameters or behavior.

Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true, but R is false
- d) A is false, but R is true

✓ **Answer:** a) Both A and R are true, and R is the correct explanation of A

5.

Assertion (A): Java is a procedural programming language.

Reason (R): Java does not support classes and objects.

Options:

- a) Both A and R are true, and R is the correct explanation of A
- b) Both A and R are true, but R is not the correct explanation of A
- c) A is true, but R is false
- d) Both A and R are false

✓ **Answer:** d) Both A and R are false

6.

Assertion (A): POP provides more modular code than OOP.

Reason (R): In POP, the entire program is divided into procedures or functions.

Options:

- a) Both A and R are true, and R is the correct explanation of A



b) Both A and R are true, but R is not the correct explanation of A

c) A is false, but R is true

d) A is true, but R is false

✓ **Answer:** c) A is false, but R is true

7.

Assertion (A): Encapsulation is the process of hiding implementation details.

Reason (R): Encapsulation restricts direct access to class members using access modifiers.

Options:

a) Both A and R are true, and R is the correct explanation of A

b) Both A and R are true, but R is not the correct explanation of A

c) A is true, but R is false

d) A is false, but R is true

✓ **Answer:** a) Both A and R are true, and R is the correct explanation of A

8.

Assertion (A): OOP makes large software systems more manageable.

Reason (R): OOP supports abstraction, encapsulation, and modularity.

Options:

a) Both A and R are true, and R is the correct explanation of A

b) Both A and R are true, but R is not the correct explanation of A

c) A is true, but R is false

d) A is false, but R is true

✓ **Answer:** a) Both A and R are true, and R is the correct explanation of A

9.

Assertion (A): Low-level languages are machine-independent and portable.

Reason (R): They use natural language for programming.

Options:

a) Both A and R are true, and R is the correct explanation of A

b) Both A and R are true, but R is not the correct explanation of A

c) A is true, but R is false

d) Both A and R are false

✓ **Answer:** d) Both A and R are false

10.

Assertion (A): POP supports inheritance, which promotes reusability.

Reason (R): Functions in POP can be called from anywhere in the program.

Options:

a) Both A and R are true, and R is the correct explanation of A

b) Both A and R are true, but R is not the correct explanation of A

c) A is false, but R is true

d) A is true, but R is false

✓ **Answer:** c) A is false, but R is true

